

Expectation-Maximization Algorithm for Gaussian Mixture Models

1 Problem Formulation and Complete Data Likelihood

1.1 Generative Model

We describe a probabilistic process that could have generated the observed data $X = \{x_i\}_{i=1}^N$. Each observation comes from one of K Gaussian components; which component produced x_i is recorded by a latent label z_i . The parameters to estimate are the mixture weights $\{\pi_k\}$, means $\{\mu_k\}$, and covariances $\{\Sigma_k\}$. Given data $X = \{x_1, \dots, x_N\}$ with $x_i \in \mathbb{R}^d$, we assume:

- Latent variables $Z = \{z_1, \dots, z_N\}$ where $z_i \in \{1, \dots, K\}$ (*one hidden label per point*).
- Prior: $p(z_i = k) = \pi_k$ with $\pi_k \geq 0$, $\sum_{k=1}^K \pi_k = 1$ (*mixture weights form a probability simplex*).
- Likelihood: $p(x_i | z_i = k) = \mathcal{N}(x_i | \mu_k, \Sigma_k)$ (*component-conditional Gaussian*).

1.2 Complete Data Likelihood

If we could observe the latent assignments Z , the joint (“complete-data”) likelihood factorizes neatly. Using indicator variables $z_{ik} = \mathbf{1}(z_i = k)$ lets us write products as powers and simplifies logs.

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$$\begin{aligned} p(X, Z | \theta) &= \prod_{i=1}^N \prod_{k=1}^K [\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)]^{z_{ik}} \\ \log p(X, Z | \theta) &= \sum_{i=1}^N \sum_{k=1}^K z_{ik} [\log \pi_k + \log \mathcal{N}(x_i | \mu_k, \Sigma_k)] \end{aligned}$$

1.3 Marginal Likelihood and the Fundamental Challenge

In reality, Z is unobserved; we must integrate it out. This creates a sum inside a log, which is the central difficulty that EM circumvents by introducing an auxiliary distribution over Z . By the law of total probability over the latent variables $Z = (z_1, \dots, z_N)$,

$$p(X | \theta) = \sum_Z p(X, Z | \theta).$$

Using conditional independence of x_i given z_i and the prior factorization of Z ,

$$p(X, Z | \theta) = \prod_{i=1}^N p(x_i, z_i | \theta) = \prod_{i=1}^N p(x_i | z_i, \theta) p(z_i | \theta).$$

Hence

$$p(X | \theta) = \sum_{z_1} \cdots \sum_{z_N} \prod_{i=1}^N p(x_i | z_i, \theta) p(z_i | \theta).$$

Because each summation is over $z_i \in \{1, \dots, K\}$ and the terms factor by i , the sums separate:

$$p(X | \theta) = \prod_{i=1}^N \sum_{z_i=1}^K p(x_i | z_i, \theta) p(z_i | \theta).$$

Now substitute the model definitions $p(z_i = k) = \pi_k$ and $p(x_i | z_i = k, \theta) = \mathcal{N}(x_i | \mu_k, \Sigma_k)$:

$$p(X | \theta) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k).$$

This establishes

$$p(X | \theta) = \sum_Z p(X, Z | \theta) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k).$$

$$\begin{aligned} p(X | \theta) &= \sum_Z p(X, Z | \theta) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k) \\ \log p(X | \theta) &= \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k) \right) \end{aligned}$$

Problem: Direct maximization is difficult due to the log-sum.

2 The EM Algorithm Framework

2.1 Evidence Lower Bound (ELBO) Derivation

We lower-bound the intractable log-likelihood by introducing any distribution $q(Z)$ over the latent variables. Jensen's inequality moves the log inside the expectation, yielding the ELBO $\mathcal{L}(q, \theta)$. EM alternates between tightening this bound (E-step) and maximizing it (M-step).

For any distribution $q(Z)$, by Jensen's inequality:

Proof.

$$\begin{aligned}
\log p(X \mid \theta) &= \log \sum_Z p(X, Z \mid \theta) \\
&= \log \sum_Z q(Z) \frac{p(X, Z \mid \theta)}{q(Z)} \\
&\geq \sum_Z q(Z) \log \frac{p(X, Z \mid \theta)}{q(Z)} \quad (\text{Jensen's inequality}) \\
&= \mathbb{E}_q[\log p(X, Z \mid \theta)] - \mathbb{E}_q[\log q(Z)] \\
&= \mathcal{L}(q, \theta)
\end{aligned}$$

□

Interpretation. The gap between $\log p(X \mid \theta)$ and $\mathcal{L}(q, \theta)$ is the KL divergence $\text{KL}(q(Z) \parallel p(Z \mid X, \theta))$. Equality holds when $q(Z) = p(Z \mid X, \theta)$, i.e., when the bound is tight.

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2.2 EM as Coordinate Ascent

At iteration t , we set $q^{(t)}(Z) = p(Z \mid X, \theta^{(t)})$ (E-step), which tightens the bound at the current $\theta^{(t)}$. Then we maximize the bound w.r.t. θ (M-step), guaranteeing non-decreasing log-likelihood.

Define the Q-function:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{p(Z \mid X, \theta^{(t)})}[\log p(X, Z \mid \theta)]$$

Theorem 2.1 (EM Monotonicity). *The EM algorithm never decreases the log-likelihood:*

$$\log p(X \mid \theta^{(t+1)}) \geq \log p(X \mid \theta^{(t)})$$

Proof.

$$\begin{aligned}
\log p(X \mid \theta^{(t+1)}) &\geq \mathcal{L}(q^{(t)}, \theta^{(t+1)}) \quad (\text{ELBO property}) \\
&\geq \mathcal{L}(q^{(t)}, \theta^{(t)}) \quad (\text{M-step maximizes Q}) \\
&= \log p(X \mid \theta^{(t)}) \quad (\text{equality at optimal q})
\end{aligned}$$

where $q^{(t)}(Z) = p(Z \mid X, \theta^{(t)})$.

□

3 E-Step: Computing Responsibilities

3.1 Posterior Distribution Factorization

Because the complete-data model factorizes over i , the posterior over Z factorizes into independent categorical distributions for each z_i given its x_i . This allows us to compute responsibilities per point.

Posterior Factorization.

$$\begin{aligned}
p(Z \mid X, \theta) &= \frac{p(X, Z \mid \theta)}{p(X \mid \theta)} \\
&= \frac{\prod_{i=1}^N \prod_{k=1}^K [\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)]^{z_{ik}}}{\prod_{i=1}^N \sum_{j=1}^K \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)} \\
&= \prod_{i=1}^N \frac{\prod_{k=1}^K [\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)]^{z_{ik}}}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)} \\
&= \prod_{i=1}^N p(z_i \mid x_i, \theta)
\end{aligned}$$

□

3.2 Responsibility Calculation

The responsibility r_{ik} is the posterior probability that point x_i belongs to component k . It is a normalized product of the prior π_k and the likelihood of x_i under component k .

Responsibility Formula.

$$\begin{aligned}
r_{ik} &= p(z_i = k \mid x_i, \theta) \\
&= \frac{p(z_i = k) p(x_i \mid z_i = k, \theta)}{p(x_i \mid \theta)} \\
&= \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}
\end{aligned}$$

□

3.3 Q-Function Derivation

Plugging the complete-data log-likelihood into the expectation under $p(Z \mid X, \theta^{(t)})$ replaces each indicator z_{ik} by its expectation r_{ik} , yielding a weighted sum where the weights are the responsibilities.

Q-Function Expression.

$$\begin{aligned}
Q(\theta \mid \theta^{(t)}) &= \mathbb{E}_{Z \mid X, \theta^{(t)}} [\log p(X, Z \mid \theta)] \\
&= \mathbb{E} \left[\sum_{i=1}^N \sum_{k=1}^K z_{ik} (\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)) \right] \\
&= \sum_{i=1}^N \sum_{k=1}^K \mathbb{E}[z_{ik}] (\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k)) \\
&= \sum_{i=1}^N \sum_{k=1}^K r_{ik} (\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \Sigma_k))
\end{aligned}$$

since $\mathbb{E}[z_{ik}] = p(z_i = k \mid x_i, \theta^{(t)}) = r_{ik}$.

□

4 M-Step: Parameter Updates

4.1 Gaussian Density Expansion

To maximize Q w.r.t. each parameter, we expand the Gaussian log-density into a constant term, a log-determinant term, and a quadratic form. Only the latter two depend on μ_k, Σ_k .

For the M-step, we need the explicit form:

$$\log \mathcal{N}(x_i | \mu_k, \Sigma_k) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k)$$

4.2 Mixing Proportions Update

We maximize Q over $\{\pi_k\}$ under the simplex constraint. A Lagrange multiplier enforces $\sum_k \pi_k = 1$, leading to a closed form: π_k equals the average responsibility mass of component k .

π_k Update. Maximize with Lagrange multiplier:

$$\mathcal{L} = \sum_{i=1}^N \sum_{k=1}^K r_{ik} \log \pi_k + \lambda \left(1 - \sum_{k=1}^K \pi_k \right)$$

Take derivative:

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^N r_{ik}}{\pi_k} - \lambda = 0 \Rightarrow \pi_k = \frac{\sum_{i=1}^N r_{ik}}{\lambda}$$

Use constraint:

$$\sum_{k=1}^K \pi_k = \frac{\sum_{k=1}^K \sum_{i=1}^N r_{ik}}{\lambda} = \frac{N}{\lambda} = 1 \Rightarrow \lambda = N$$

Thus:

$$\pi_k = \frac{1}{N} \sum_{i=1}^N r_{ik}$$

□

4.3 Means Update

Holding Σ_k fixed, Q as a function of μ_k is a weighted (by r_{ik}) sum of quadratic forms. Setting the gradient to zero yields the familiar responsibility-weighted mean.

μ_k Update. The relevant terms in Q are:

$$Q_{\mu_k} = -\frac{1}{2} \sum_{i=1}^N r_{ik} (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k)$$

Take derivative using matrix calculus:

$$\begin{aligned}\frac{\partial Q_{\mu_k}}{\partial \mu_k} &= -\frac{1}{2} \sum_{i=1}^N r_{ik} \left(-2 \Sigma_k^{-1} (x_i - \mu_k) \right) \\ &= \sum_{i=1}^N r_{ik} \Sigma_k^{-1} (x_i - \mu_k)\end{aligned}$$

Set to zero:

$$\sum_{i=1}^N r_{ik} \Sigma_k^{-1} (x_i - \mu_k) = 0 \Rightarrow \sum_{i=1}^N r_{ik} x_i = \mu_k \sum_{i=1}^N r_{ik}$$

Thus:

$$\mu_k = \frac{\sum_{i=1}^N r_{ik} x_i}{\sum_{i=1}^N r_{ik}}$$

□

4.4 Covariance Matrix Update

Maximizing over Σ_k balances two forces: increasing $\log |\Sigma_k|$ penalizes spread, while matching the weighted scatter of points around μ_k encourages appropriate variance. Taking the derivative w.r.t. Σ_k and setting to zero yields the standard weighted covariance.

Σ_k Update. The relevant terms:

$$Q_{\Sigma_k} = -\frac{1}{2} \sum_{i=1}^N r_{ik} \left[\log |\Sigma_k| + (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k) \right]$$

Use matrix derivative identities:

$$\begin{aligned}\frac{\partial \log |\Sigma_k|}{\partial \Sigma_k} &= \Sigma_k^{-1} \\ \frac{\partial (x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k)}{\partial \Sigma_k} &= -\Sigma_k^{-1} (x_i - \mu_k) (x_i - \mu_k)^\top \Sigma_k^{-1}\end{aligned}$$

Thus:

$$\begin{aligned}\frac{\partial Q_{\Sigma_k}}{\partial \Sigma_k} &= -\frac{1}{2} \sum_{i=1}^N r_{ik} \left[\Sigma_k^{-1} - \Sigma_k^{-1} (x_i - \mu_k) (x_i - \mu_k)^\top \Sigma_k^{-1} \right] \\ &= -\frac{1}{2} \Sigma_k^{-1} \left[\sum_{i=1}^N r_{ik} I - \sum_{i=1}^N r_{ik} (x_i - \mu_k) (x_i - \mu_k)^\top \Sigma_k^{-1} \right]\end{aligned}$$

Set to zero and multiply by Σ_k on both sides:

$$\sum_{i=1}^N r_{ik} I = \sum_{i=1}^N r_{ik} (x_i - \mu_k) (x_i - \mu_k)^\top \Sigma_k^{-1}$$

Multiply by Σ_k :

$$\left(\sum_{i=1}^N r_{ik}\right) \Sigma_k = \sum_{i=1}^N r_{ik} (x_i - \mu_k)(x_i - \mu_k)^\top$$

Thus:

$$\Sigma_k = \frac{\sum_{i=1}^N r_{ik} (x_i - \mu_k)(x_i - \mu_k)^\top}{\sum_{i=1}^N r_{ik}}$$

□

5 Complete EM Algorithm

5.1 Algorithm Summary

The algorithm alternates between computing soft assignments (E-step) and re-estimating parameters by responsibility-weighted maximum likelihood (M-step). The effective count for cluster k is $N_k = \sum_i r_{ik}$.

1. **Initialize:** $\theta^{(0)} = \{\pi_k^{(0)}, \mu_k^{(0)}, \Sigma_k^{(0)}\}_{k=1}^K$ (e.g., via k -means or random).
2. **Repeat until convergence:**
 - **E-step:** For $i = 1, \dots, N$, $k = 1, \dots, K$:

$$r_{ik} = \frac{\pi_k^{(t)} \mathcal{N}(x_i | \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^K \pi_j^{(t)} \mathcal{N}(x_i | \mu_j^{(t)}, \Sigma_j^{(t)})}$$

(Compute posterior responsibilities for each point and component.)

- **M-step:** For $k = 1, \dots, K$:

$$\begin{aligned} N_k &= \sum_{i=1}^N r_{ik} \\ \pi_k^{(t+1)} &= \frac{N_k}{N} \\ \mu_k^{(t+1)} &= \frac{1}{N_k} \sum_{i=1}^N r_{ik} x_i \\ \Sigma_k^{(t+1)} &= \frac{1}{N_k} \sum_{i=1}^N r_{ik} (x_i - \mu_k^{(t+1)})(x_i - \mu_k^{(t+1)})^\top \end{aligned}$$

(Update mixture weight, mean, and covariance using soft counts.)

5.2 Convergence Monitoring

Monitoring the data log-likelihood ensures EM makes progress and allows a principled stopping rule. Because EM is monotone, the sequence is non-decreasing and typically stabilizes.

Compute log-likelihood:

$$\log p(X \mid \theta^{(t)}) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k^{(t)} \mathcal{N}(x_i \mid \mu_k^{(t)}, \Sigma_k^{(t)}) \right)$$

Stop when $|\log p(X \mid \theta^{(t+1)}) - \log p(X \mid \theta^{(t)})| < \epsilon$.

6 Theoretical Properties

EM is a bound-optimization (majorize–maximize) method. Under mild conditions, it converges to a stationary point of the likelihood. The monotonicity is guaranteed by the ELBO argument, but the limit may be a local optimum (or a saddle), so initialization matters.

Theorem 6.1 (EM Convergence). *Under mild regularity conditions:*

1. *The sequence $\{\theta^{(t)}\}$ converges to a stationary point of $\log p(X \mid \theta)$*
2. *$\log p(X \mid \theta^{(t)})$ converges monotonically*
3. *The convergence rate is linear near the optimum*

Sketch. 1. The EM algorithm is a generalized gradient ascent method

2. Monotonicity follows from the ELBO property
3. Convergence to stationary points follows from Zangwill’s global convergence theorem
4. Linear convergence rate can be shown using the matrix of the EM operator

□

Practical Notes. (i) *Degeneracy.* If a covariance collapses, the likelihood can blow up; regularize Σ_k or enforce minimum eigenvalues. (ii) *Label switching.* Component indices are not identifiable; interpret clusters up to permutation. (iii) *Initialization.* Good starts (e.g., k -means) mitigate poor local optima. (iv) *Diagonal/Shared covariances.* Constraining Σ_k can improve robustness and reduce variance in high dimensions.